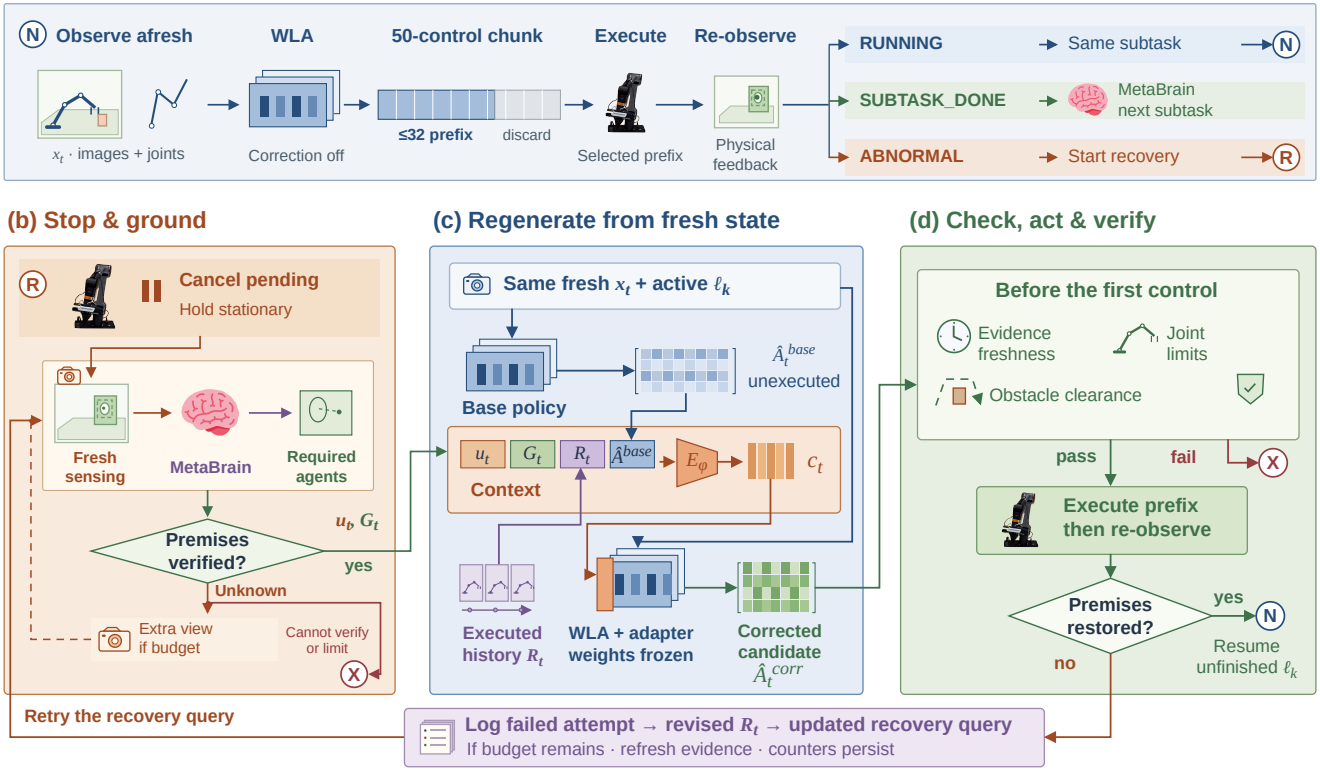

(a) Nominal execution · active subtask ℓ_k
N Observe afresh
WLA
50-control chunk
Execute
Re-observe
x_t · images + joints
Correction off
≤32 prefix
discard
Selected prefix
Physical feedback
RUNNING
Same subtask
N
SUBTASK_DONE
MetaBrain next subtask
ABNORMAL
Start recovery
R
(b) Stop & ground
R
Cancel pending
Hold stationary
Fresh sensing
MetaBrain
Required agents
u_t, G_t
Premises verified?
yes
Unknown
Extra view if budget
Cannot verify or limit
X
Retry the recovery query
(c) Regenerate from fresh state
Same fresh x_t + active ℓ_k
Base policy
$\hat{A}_t^{base}$ unexecuted
u_t G_t R_t $\hat{A}^{base}$ E_φ c_t
Context
Executed history R_t
WLA + adapter weights frozen
Corrected candidate $\hat{A}_t^{corr}$
(d) Check, act & verify
Before the first control
Evidence freshness
Joint limits
Obstacle clearance
pass
fail
X
Execute prefix then re-observe
Premises restored?
yes
N
Resume unfinished ℓ_k
no
Log failed attempt → revised R_t → updated recovery query
If budget remains · refresh evidence · counters persist